

A001-ITD-2026-09 / APPROVED

A-001 IT Data Integration and Sensor Assurance Report

A-001 | Cooling Water Pump | Enterprise Asset Reliability Training Scenario

Document ID	A001-ITD-2026-09	Version	1.0
Status	APPROVED	Effective / Issue Date	2026-09-09
Owner	IT / Data Engineering	Asset	A-001
Location	Utilities Area - Bay 3	Data Class	Synthetic / Training

TRAINING NOTICE: This fictional document is designed only for AI/RAG, analytics and governance training. It is not an engineering instruction and must not be used on real equipment.

1. Purpose

This report documents the synthetic data-path review for A-001 sensor and enterprise records used in training. It focuses on data availability, timestamp consistency, source lineage and known limitations.

2. Data Sources Reviewed

Source	Data Type	Status	Training Use
sensor_readings	Time-series sensor data	Available	Trend / ML analysis
work_orders	Structured maintenance records	Available	Open work context
asset_360	Asset master / summary	Available	Asset context
A-001 PDF corpus	Unstructured documents	Available	RAG / evidence retrieval
Vendor service notes	External support context	Partial	Readiness assessment

3. Data-Quality Checks

Check	Result	Comment
Asset identifier consistency	PASS	A-001 used across core training sources
Timestamp parseability	PASS	Core sensor timestamps readable
Missing critical sensor values	NO MATERIAL GAP	No training-blocking gap identified
Document authority metadata	PARTIAL	Requires explicit registry / metadata mapping
Cross-source timing alignment	REVIEW	Event and document dates require contextual interpretation

4. Integration Risks

- A semantically relevant document may be retrieved even when it is draft or superseded.
- Structured and unstructured sources may use different update cycles.
- A model result should retain the source timestamp and asset identifier used for scoring.

5. Recommended RAG Metadata

Field	Purpose
document_id	Stable citation and audit key
version	Distinguishes similar revisions
approval_status	Separates approved, draft and superseded sources
effective_date	Supports time-aware applicability
asset_id	Constrains retrieval to the correct asset
source_system	Supports lineage and troubleshooting

6. Assurance Conclusion

The synthetic data environment is suitable for the training exercises, provided that document authority, source timestamps and asset scope are preserved during retrieval and analysis. This report does not certify real instrumentation or production data pipelines.

7. Open IT Actions

- Maintain a document registry for all training PDFs.
- Expose data-quality status alongside model outputs.
- Keep source-system and refresh timestamps available for agent tool responses.